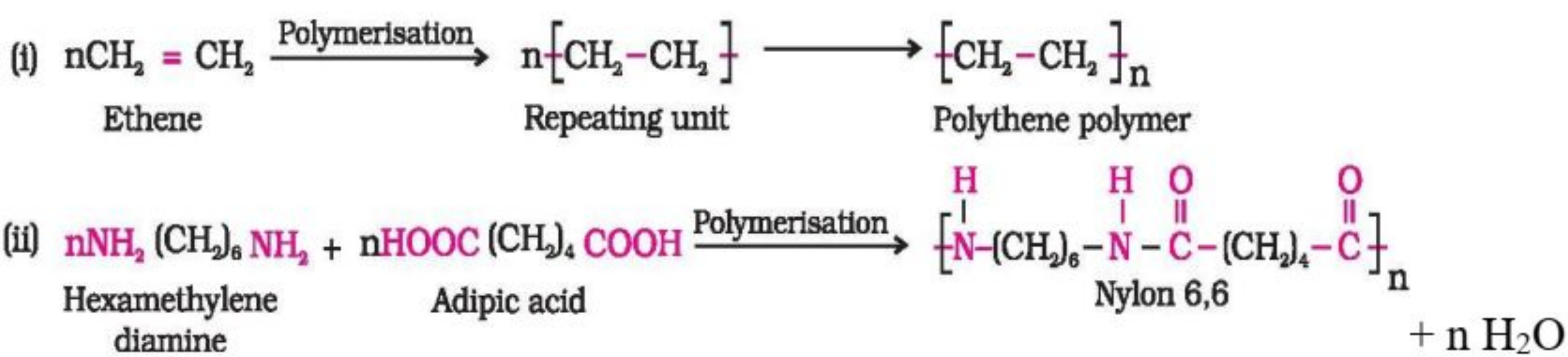
Module IV

Polymers

- Polymerization is a type of chemical reaction in which two or more molecules combine together with or without the evolution of heat or elimination of water to form a molecule of high molecular weight. Thus the starting material is called monomer and the resulting product is called polymer
- the starting material is called monomer and the resulting product is called polymer.



Classification of Polymers

Basis of Classification	Polymer Type
Origin	Natural, Semi synthetic, Synthetic
Thermal Response	Thermoplastic, Thermosetting
Mode of Formation	Addition, Condensation
Geometric Structure	Linear, Branched, Cross-linked
Applications and Physical Properties	Rubber, Plastic, Fibers
Tacticity	Isotactic, Syndiotactic, Atactic
Crystallinity	Non crystalline (or amorphous), Semi-crystalline, Crystalline

A. Origin

Based on their occurrence (or lack of) in nature, polymers can be classified into three types:

i) Natural Polymers

Eg: natural rubber, natural silk, wool, DNA, cellulose, starch and proteins

ii) Semi synthetic Polymers

Eg: hydrogenated natural rubber, cellulosic, cellulose nitrate, methyl cellulose, etc.

iii)Synthetic Polymers

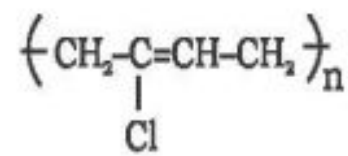
Eg: polyvinyl alcohol, polyethylene, polystyrene, polysulfone, etc.

B. Thermal Response

On the basis of temperature variation, polymers can be differentiated into two groups:

i) Thermoplastic polymers:

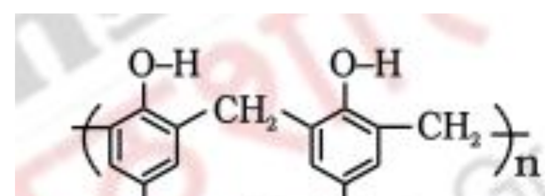
Polymers which get softened or plasticized repeatedly on application of thermal energy, without considerable change in properties if treated with certain precautions. Eg: Neoprene



Neoprene

ii) Thermosetting polymers:

Polymers which undergo certain variations on heating and get converted into an infusible mass are known as thermosetting polymers.



Bakelite

C. Mode of Formstion

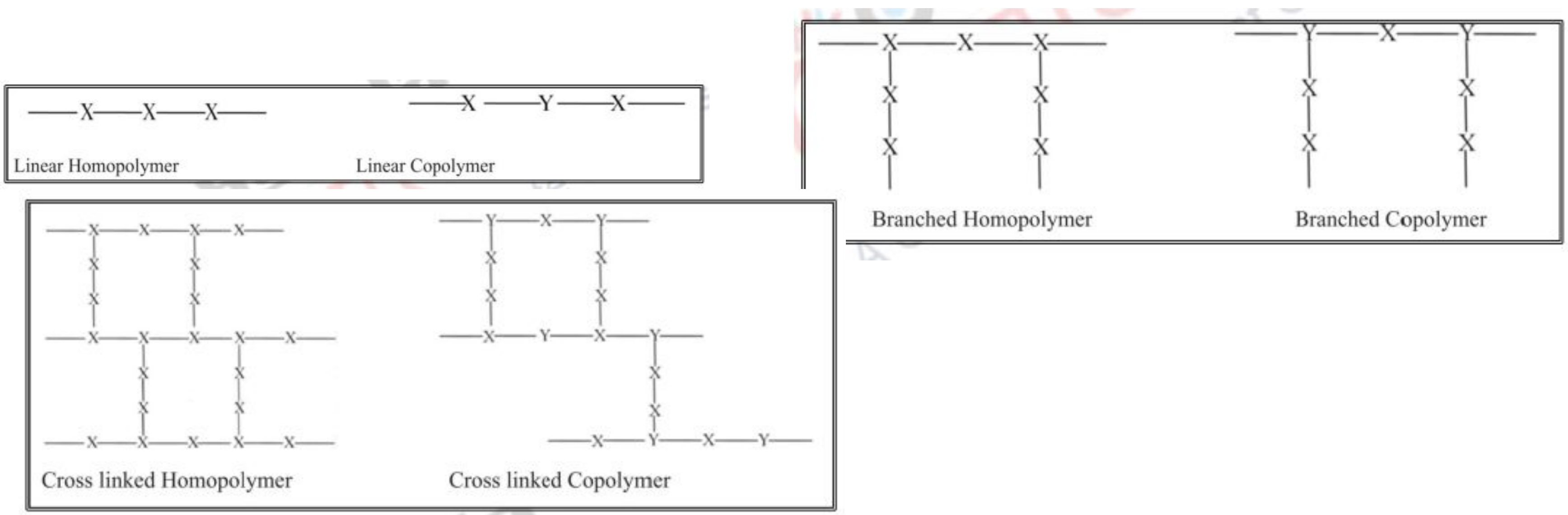
Based on the reaction involved in their formation, polymers can be categorized as:

i) Addition polymers:

$$n\,CH_2=CH_2 \longrightarrow -\!\!\left(CH_2-CH_2\right)\!\!-_n \quad \text{Homopolymer}$$

Ethene → Polythene

ii) Condensation polymers:



Application and Physical Properties

Depending on its ultimate use and form, a polymer can be differentiated as:

- Rubber (Elastomers)
- Plastics
- Fibers

LUBRICANTS

- A lubricant is substance (often liquid) introduced between two moving surfaces to reduce the friction between them.
- Fluid which is introduced in between moving parts in order to reduce the friction, generated heat & wear and tear of machine parts are called Lubricants.
- This process of introducing lubricant is called Lubrication.

COMPOSITION

- Typically contains 90% base oil(petroleum-mineral oils) and less than 10% additives
- Non liquid lubricants contains Grease, powder(dry), plumbing etc., graphite, Molybdenum disulphite, Teflon tape used in plumbing etc
- Those non liquid lubricants provide lubrication at higher temp. (up to 350°C).

ADDITIVES USED IN LUBRICANTS

(1)Anti oxidant --- Aromatic amines,Phenols,Sulphides and phosphates

(2)Corrosion Inhibitor --- Amino salts and salts of sulphonic acids

(3) Antiwear agents --- Tricresyl phosphate

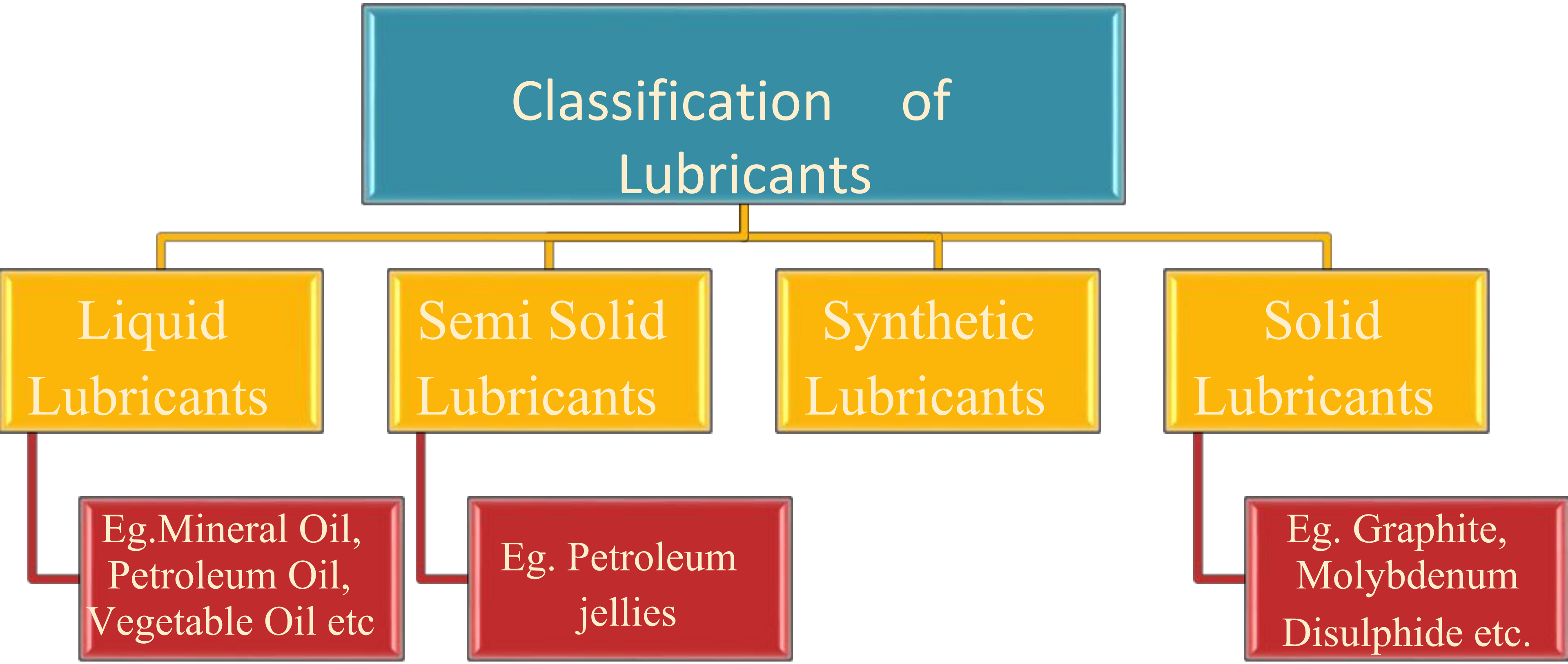
(4) Foam inhibitors --- Glycerols

Functions of Lubricants

- ✔ It acts as a thermal barrier and reduces friction and wear and prevents welded joints
- ✔ Avoids seizure of moving surfaces
- ✔ Acts as coolant
- ✔ Acts as a seal and prevents entry of dust, moisture, and dirt between moving parts
- ✔ Some lubricants acts as corrosion inhibitors thus reduce operational cost .

Characteristics

- High boiling point
- Adequate Viscosity
- Low freezing point
- High oxidation resist
- Non Corrosive properties
- Good thermal stability


Classification of Lubricants
Liquid Lubricants
Semi Solid Lubricants
Synthetic Lubricants
Solid Lubricants
Eg.Mineral Oil, Petroleum Oil, Vegetable Oil etc
Eg. Petroleum jellies
Eg. Graphite, Molybdenum Disulphide etc.

Liquid Lubricants:

It includes animal oils, vegetable oils, petroleum oils, synthetic lubricants.

Animal oils: tallow oil, whale oil etc.

Vegetable oils: castor oil, palm oil etc

Petroleum oils: petroleum fractions

Synthetic lubricants: polyglycol, silicones etc.

Semi-solid Lubricants (Grease):

Semi-solid Lubricants are formed by emulsifying oil and fat with thickening agents like soap of sodium, calcium, lithium, aluminum at higher temperature.

Classification

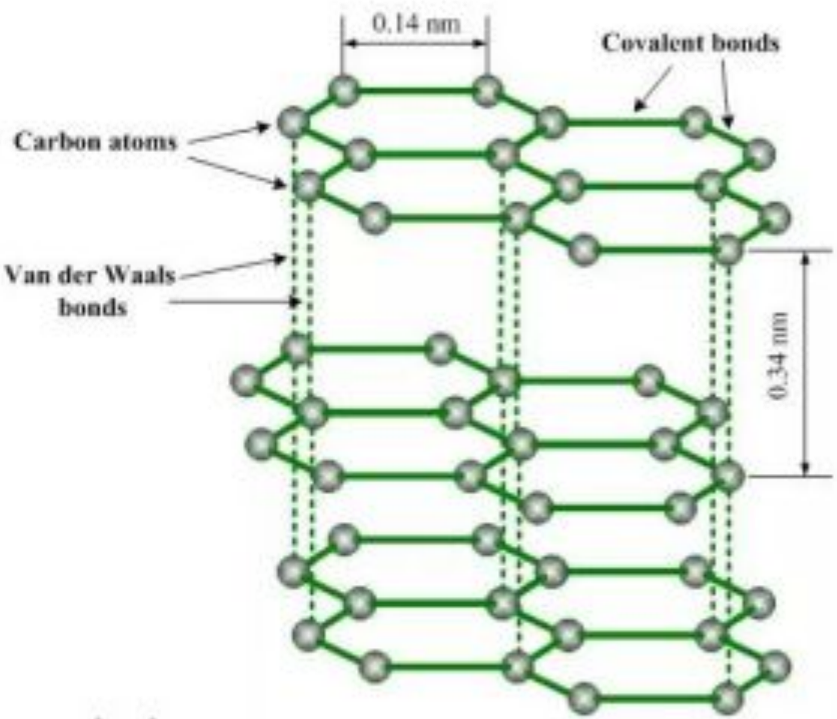
Soda based: In this case sodium soaps are used as a thickening agent in mineral or petroleum oil. They are slightly soluble in water. They can be used up to 175°C.

Lithium based: In this case lithium soaps are emulsifying with petroleum oil. They are water resistance and used up to 15°C.

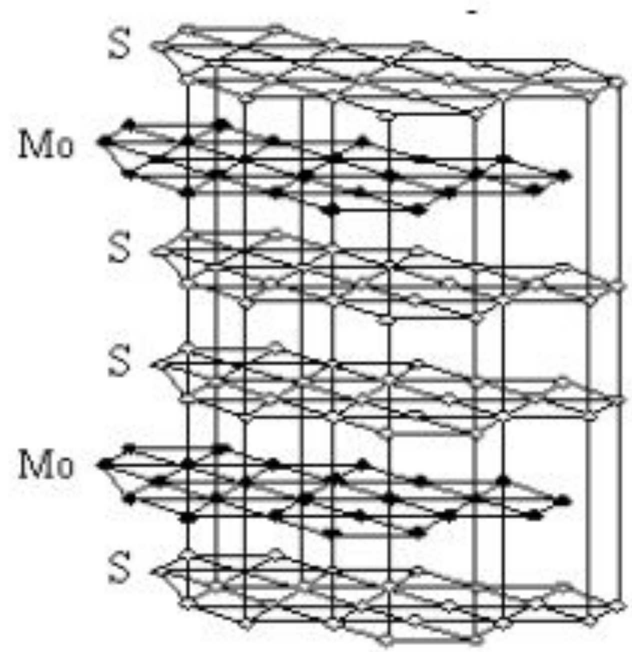
Calcium based: In this case calcium soaps are emulsifying with petroleum oil. They are also water resistant and used up to 80°C. At higher temperature soap and petroleum oil are separate from each other.

Solid Lubricants:

Graphite, molybdenum disulphide (MoS_2), boron nitride $(BN)_x$ are predominantly used as a solid lubricants. They are used under high temperature and high load (pressure).



Graphite structure



Molybdenum disulphide structure

Viscosity

- It’s a measure of a fluid’s resistance to flow.
- Viscosity of the lubricating oil determines its performance under operating conditions.
- A low viscosity oil is thin and flows easily .
- A high viscosity oil is thick and flows slowly.
- As oil heats up it becomes more viscous (*Becomes thin)*
- Too low viscosity of the liquid > Lubricant film cannot be maintained between the moving surfaces > Excessive wear.
- Too high viscosity of the liquid > Excessive friction.
- Selected Lubricant must be proper viscous.
- Viscosity is usually expressed in centipoise or centistoke.

DETERMINATION OF VISCOSITY INDEX:

- **Test oil is compared with two standard oils with viscosity index 0 & 100**
- **Oil with viscosity index =0(Gulf oil) (L) naphthenic oil series**
- **Oil with viscosity index =100(Pennsylvanian oil) (H)**

$$\boxed{V.I = \frac{L-U}{L-H} \times 100}$$

Iodine Number

- Iodine number is the number of Gms equivalent of iodine to amount of ICl absorbed by 100gm of oil.
- Each oil has its specific Iodine Number.
- So Iodine Number determines the extent of contamination of oil.
- Low Iodine Number is desirable in oils.

Some oils and their Iodine Numbers are given below :

Iodine Number	Oil	Example
>150	Drying oil	Linseed oil, tung oil
100-150	Semidrying oil	Castor oil , Soyabean oil
<100	Non-Drying oil	Coconut oil, Olive oil

The temperature at which separation of the two phases (Aniline + oil) takes place is the Aniline Point.

Flash point and Fire point

- Flash Point is the min temp at which the lubricant vaporizes that ignite for a moment when tiny flame is brought near.

  5 seconds when tiny flame is brought

- Fire Point is the Min temp at which the lubricant's vapours burn constantly for
- If flash point < 140°F = Flammable liquids

  And if flash point > 140°F =Combustible liquids.

Refractories

- Materials that can withstand high temp without softening and deformation in their shape.
- Used for the construction of furnaces, converters, kilns, crucibles, ladles etc.

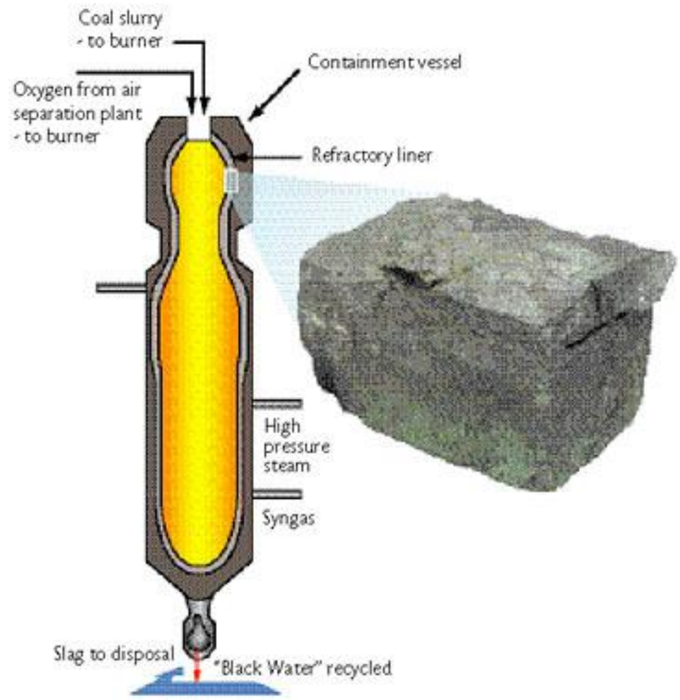


Fig (1) : Schematic of gasifier cross section showing the location of the spent refractory brick

Refractories Examples

Fig (2 & 3) : Various shapes of materials used in refractories

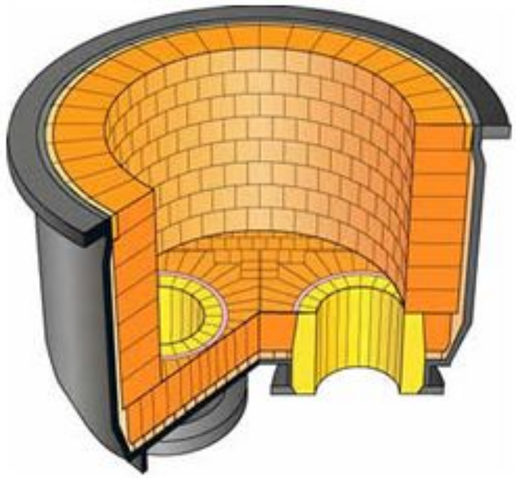


Fig (4 & 5) : Furnace inner lining showing refractory materials

How is it manufactured and used

Characteristics of Refractories

- Infusible at operating temperature
- Chemically inert towards corrosive gases, liquids etc.
- Should not suffer change in size at operating temp
- Should have high refractoriness
- Should have high load bearing capacity at operating temp.

Classification of Refractories

Refractories

CHEMICAL NATURE

- Acidic Refractories
  - Silica , Alumina
- Basic Refractories
  - Magnesite, Dolomite
- Neutral Refractories
  - Graphite, Carborundun

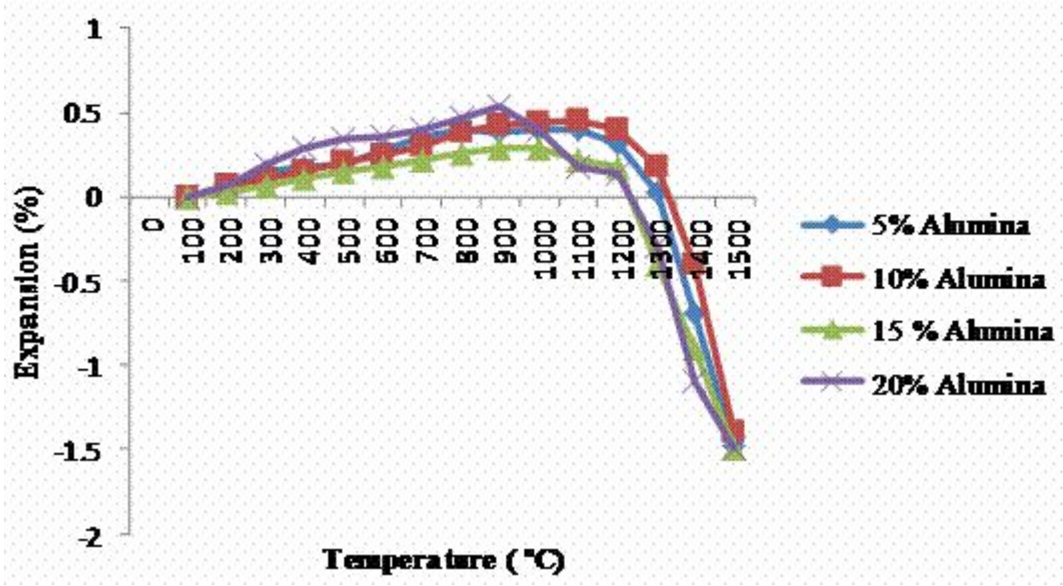
REFRACTORINESS

- Low Heat Duty
- Intermediate Heat Duty
- High Heat Duty
- Super Heat Duty

Properties of Refractories

Refractoriness : " It is the ability to withstand very high temp. without softening or deformation under particular service condition. "

The below figure shows refractoriness of Alumina at various temperatures



Properties of Refractories contd..

- Most of the refractories are mixtures of several metallic oxides and they don't have a sharp melting point
- Refractoriness of a refractory is generally measured as the softening temperature and is expressed in terms of Pyrometric Cone equivalent (PCE)

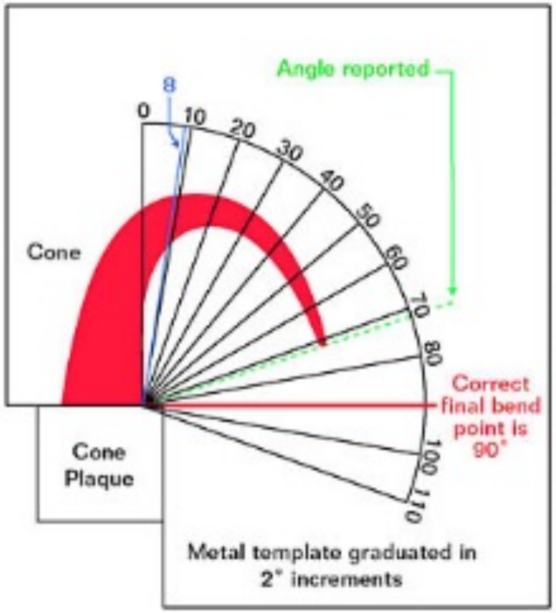
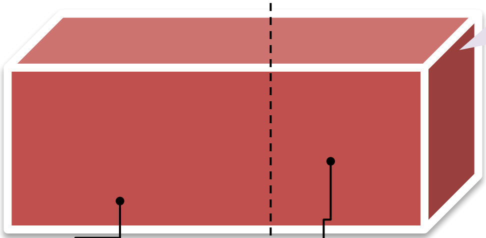


Figure showing softening temperature of a cone at different time intervals

Acidic Refractory : Alumina Bricks or fire clay bricks

Composition of Alumina Bricks



50 % or more of Calcium Bauxite Al_2O_3 + Silica,Grog (fire clay)

+ WATER

PASTE

Moulding: Machine pressing or slip casting

Drying and Firing at 1200°C to 1400°C (6 to 8 days)

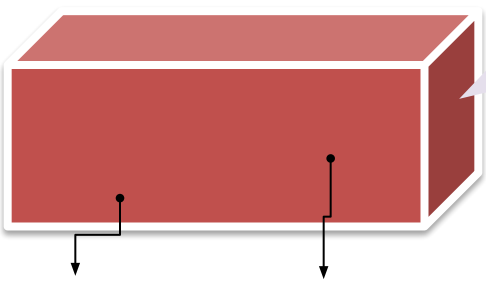
PROPERTIES

- Low coefficient of expansion
- High porosity
- Little tendency to thermal spalling
- Resistance to slag
- Stable and wear resistance
- High temperature load bearing capacity
- Inert to gases lik COl_2 , H_2 and natural gas

- **Medium Duty Bricks :** 50 – 60 % of Al_2O_3
  - Cement rotary klins
  - Soaking pits
  - Reheating furnace
  - Hearths & walls
- **High Duty Bricks** : 75 % of Al_2O_3
  - Hottest zones of cement rotary klins
  - Brass melting reverberatories
  - Aluminium melting furnaces

Basic Refractory : Magnesite bricks

Composition of Magnesite Bricks



more of Calcined Magnsite MgO + Castic Magnesia or Iron Oxide, Sulphite Lye

+ **WATER**

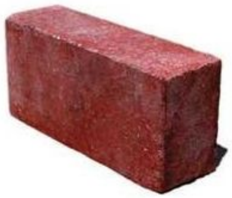
PASTE

Moulding: Machine pressing

Drying and Firing at 1500^{o}C (8 hours)

PROPERTIES

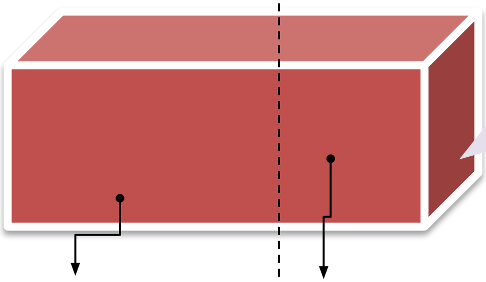
- Withstands 2000 ^{o}C without load and upto 1500 ^{o}C under load of 3.5Kg / cm^2
- Good resistance to basic slags
- Little shrinkage and more of spalling
- Poor resistance to abrasion
- Easily combines with Carbon dioxide and water
- Highly sensitive to sudden change to temperature



- **Steel Industry for lining of**
  - Basic Convertors
  - Open – hearth furnace
- **Copper Convertors**
- **Reverberatory furnaces**
- **Refining furnaces for Gold, silver and platinum etc.,**
- **Hot mixture linings**

Neutral Refractory : Caborundum or Silicon Carbide (SiC) Bricks

Composition of Silicon Carbide Bricks



60 % of Sand SiO_2 + 40 % Coke (C)

+ SAW DUST, WATER

PASTE

↓

Moulding: Machine pressing

↓

Drying and Firing at 1500°C

PROPERTIES

- High thermal conductivity
- Low thermal expansion
- ‘Clay bonded’ can be used upto **1750°C**
- ‘Silicon nitrate bonded’ poses high strength and thermal shock resistance
- ‘Self bonded’ has high refractoriness, strength , density, abrasion resistance and chemical resistance
- Oxidizes when heated at **900°C -1000°C** , which can be prevented by coating with zirconium

- **Partial Walls of**
  - Chamber klins, coke ovens, muffle furnace
- **Floor for**
  - Heat treatment furnace
- **Heating elements in forms of**
  - Rods and Bars (globars)